Hall Ticket Number:

MR-21

Code: B415IT5S2

MAHATMA GANDHI INSTITUTE OF TECHNOLOGY (Autonomous)

B.Tech. IV Semester Supplementary Examinations, December 2025/January 2026

Discrete Mathematics

(Information Technology)

Time: 3 Hours | ***Answer ALL Questions*** | Max. Marks: 70

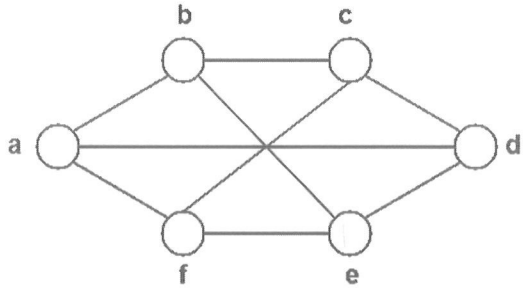
Part-A (10 x 2 = 20 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
1. a)	Give the symbolic form of the following: If either Jerry takes calculus or Ken takes sociology then Larry will take English.	2	1	1	1,2
1. b)	Write the converse and contra positive of the statement: "If two angels in triangle are equal, then triangles are isosceles".	2	1	1	1,2
1. c)	Define relation. Give an example.	2	1	2	1
1. d)	What is an equivalence relation? Give an example.	2	1	2	1
1. e)	How many 5 letter words are there where the first and last letters are vowels? (Repetition is allowed)	2	1	3	1,2
1. f)	In how many ways can we draw a heart or a king from a pack of cards?	2	1	3	1,2
1. g)	Solve the following recurrence relation by substitution method. $a_n=a_{n-1}+ n$ where $a_0=2$.	2	3	4	1,2
1. h)	Find the generating function for the sequence of numbers 1,-2,3,-4,--	2	1	4	1,2
1. i)	Define degree sequence of a graph. Give an example.	2	1	5	1
1. j)	Define isomorphism. Give an example.	2	1	5	1

Part-B (5 x 10 = 50 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
2. a)	Show that R→S can be derived from the premises P→(Q→S), ⅂RVP and Q.	5	2	1	1,2
2. b)	Show that the following premises are inconsistent. i. If jack misses many classes through illness then he fails high school. ii. If jack fails high school, then he is uneducated. iii. If jack reads a lot of books then he is not uneducated. iv. Jack misses many classes through illness and reads a lot of books.	5	2	1	1,2
	OR				
2. c)	Show that $(x)(P(x) \rightarrow Q(x)) \wedge (x)(Q(x) \rightarrow R(x)) \Rightarrow (x)(P(x) \rightarrow R(x))$.	5	2	1	1,2
2. d)	Show that the following statements are logically equivalent without using truth table. $(P \rightarrow R) \wedge (Q \rightarrow R) \Leftrightarrow (P \vee Q) \rightarrow R$.	5	2	1	1,2
3. a)	Let A={1, 2, 3, 4, 6, 8, 12, 24} and the relation ' $\leq$ ' be such that $x \leq y$ if x divides y. Draw Hasse diagram of $<A, \leq>$.	5	3	2	1,2
3. b)	Let X={1,2,3} and f, g, h be functions from X to X given by f={<1,2>, <2,3>,<3,1>} g={<1,2>,<2,1>,<3,3>} h={<1,1>,<2,2>,<3,1>} Find whether f o(g o h)=(f o g) o h	5	3	2	1,2
	OR				
3. c)	What is a binary relation? Explain the properties of binary relations with examples.	5	2	2	1
3. d)	What is a POSET? Draw the Hasse diagram representing the positive divisors of 36.	5	2	2	1,2

P.T.O.

4. a)	In how many ways can 10 people be seated in a row so that certain pair of them are not next to each other?	5	4	3	1,2
4. b)	State and prove pigeon-hole principle.	5	2	2	1
	OR				
4. c)	State and prove Binomial theorem. Expand $(5a+7b)^8$ using binomial theorem.	5	2	2	1,2
4. d)	In how many ways can a committee of 5 ladies and 4 gents be chosen from 9 ladies and 15 gents, if gent A refuses to take part if lady B is on the committee?	5	4	3	1,2
5. a)	Find a general expression for a solution to the recurrence relation, $a_n-5a_{n-1}+6a_{n-2}= n(n-1)$ for $n>=2$	5	3	4	1,2
5. b)	Solve $a_n-6a_{n-1}+12a_{n-2}-8a_{n-3} = 0$ for $n>=3$ by the method of characteristic roots.	5	3	4	1,2
	OR				
5. c)	Find a general expression for a solution to the recurrence relation $a_n-8a_{n-1}+12a_{n-2} - 0$ for $n>=2$.	5	3	4	1,2
5. d)	Solve $a_n-7a_{n-1+}10a_{n-2} = 0$, for $n>=2$, given $a_0=10$, $a_1=41$ using generating functions method.	5	3	4	1,2
6. a)	Define Planar graph. Is $K_{3,3}$ a planar graph? Justify your answer.	5	2	5	1
6. b)	Explain the steps involved in deriving a spanning tree using Breadth first search algorithm with a suitable example.	5	2	5	1
	OR				
6. c)	Define chromatic number of a graph. Is the following graph 2-chromatic? Justify.  	5	2	5	1
6. d)	Prove that if G is a connected planar graph, then $\|V\| - \|E\| + \|R\| = 2$ where $\|R\|$ denote number of regions, $\|E\|$ denote number of edges and $\|V\|$ the number of vertices.	5	4	5	1

M: Marks; L: Bloom's Taxonomy Level; CO: Course Outcome; PO: Programme Outcome